Fourier Multiplication Circuit

Discovered in a 1-Layer Transformer Trained on $(a+b) \bmod 113$

Auto-generated by Grokking Circuit Discovery Tool

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Circuit Summary

Task: $(a+b) \bmod 113$ **FVE:** 0.9580 **Formula Accuracy:** 100.00%
Key frequencies: $\mathcal{K} = \{3, 38, 51\}$ (3 frequencies)
Neurons assigned: 2/512 ($k=3$: 1, $k=38$: 1, $k=51$: 0)

The Core Equation

$$\boxed{\text{Logit}(c \mid a,b) = \sum_{k\in\mathcal{K}} \underbrace{\alpha_k}_{\substack{\text{learned}\\\text{amplitude}}} \cdot \underbrace{\cos\left(\frac{2\pi k(a+b-c)}{113}\right)}_{\substack{\text{max at } c=(a+b) \bmod 113\\\text{(constructive interference)}}}} \quad (1)$$

Data Flow Pipeline

$$\underbrace{a,b}_{\text{inputs}} \xrightarrow{W_E} \underbrace{\cos(\omega_k t),\ \sin(\omega_k t)}_{\text{Fourier embedding}} \xrightarrow{\text{Attn}} \underbrace{\text{degree-2 products}}_{\text{attn wt.}\times\text{OV}}$$

$$\xrightarrow{\text{MLP}} \underbrace{\cos(\omega_k(a+b)),\ \sin(\omega_k(a+b))}_{\text{trig addition identities}} \xrightarrow{W_U} \underbrace{\cos(\omega_k(a+b-c))}_{\text{logits}}$$

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1 Discovered Circuit: Fourier Multiplication Algorithm

The model computes $(a+b) \mod 113$ using 3 key frequencies $k \in \{3, 38, 51\}$ with angular frequencies $\omega_k = \frac{2\pi k}{113}$.

1.1 Step 1: Embedding (Token → Fourier Components)

The embedding matrix W_E maps each one-hot token $t \in \{0, \ldots, 112\}$ onto sinusoidal components:

$$\mathbf{x}_a^{(0)} = \underbrace{W_E \cdot \mathbf{e}_a}_{\text{token embedding of } a} + \underbrace{\mathbf{p}_0}_{\text{positional embedding (pos 0)}} \tag{2}$$

$$\approx \sum_{k \in \mathcal{K}} \left[\underbrace{\alpha_k \cos\left(\frac{2\pi k \cdot a}{113}\right)}_{\substack{\text{cosine component} \\ \text{from } W_E \text{ row } a}} \cdot \mathbf{u}_k^{(\cos)} + \underbrace{\beta_k \sin\left(\frac{2\pi k \cdot a}{113}\right)}_{\substack{\text{sine component} \\ \text{from } W_E \text{ row } a}} \cdot \mathbf{u}_k^{(\sin)} \right] \tag{3}$$

where $\mathcal{K} = \{3, 38, 51\}$ are the key frequencies discovered via Fourier analysis of W_E.

Similarly for token b at position 1:

$$\mathbf{x}_b^{(0)} = \underbrace{W_E \cdot \mathbf{e}_b}_{\text{token embedding of } b} + \underbrace{\mathbf{p}_1}_{\text{positional embedding (pos 1)}} \tag{4}$$

1.2 Step 2: Attention (Move Fourier Info to Output Position)

The attention heads compute scores from the '=' token (position 2) to positions 0 and 1, then produce a weighted combination:

$$\text{score}_{= \to a}^{(j)} = \frac{\overbrace{\mathbf{x}_=^{(0)} \cdot W_Q^{(j)}}^{\text{query from '=' position}} \cdot \overbrace{\left(W_K^{(j)}\right)^\top \cdot \mathbf{x}_a^{(0)}}^{\text{key from position } a}}{\underbrace{\sqrt{d_{\text{head}}}}_{\text{scaling factor}}} \tag{5}$$

$$A_0^{(j)} = \underbrace{\sigma\left(\text{score}_{= \to a}^{(j)} - \text{score}_{= \to b}^{(j)}\right)}_{\substack{\text{softmax over 2 elements} \\ = \text{sigmoid of difference}}} \tag{6}$$

$$\approx \underbrace{0.5}_{\text{uniform baseline}} + \underbrace{\gamma_j \left(\cos\left(\omega_{k_j} \cdot a\right) - \cos\left(\omega_{k_j} \cdot b\right) \right)}_{\substack{\text{periodic modulation at frequency } k_j \\ (\text{from } C^{(j)} = W_E^\top W_K^{(j)\top} W_Q^{(j)} \mathbf{x}_=^{(0)})}} \tag{7}$$

$$\text{attn\_out}^{(j)} = \underbrace{A_0^{(j)}}_{\text{attn weight to } a} \cdot \overbrace{W_O^{(j)} W_V^{(j)} \mathbf{x}_a^{(0)}}^{\text{OV circuit applied to } a} + \underbrace{A_1^{(j)}}_{\text{attn weight to } b} \cdot \overbrace{W_O^{(j)} W_V^{(j)} \mathbf{x}_b^{(0)}}^{\text{OV circuit applied to } b} \tag{8}$$

Key insight: Since $A_0^{(j)}$ is approximately linear in $\cos(\omega_{k_j} a)$ and the OV circuit outputs $\cos(\omega_{k_j} a)$, their product creates *degree-2 polynomials* of sines and cosines — exactly what is needed for the trig identity in the next step.

1.3 Step 3: MLP (Compute Trig Identities)

The residual stream after attention contains degree-2 products. The MLP neurons compute the key trigonometric identities:

$$\mathbf{x}^{(1)} = \underbrace{\mathbf{x}_{=}^{(0)}}_{\substack{\text{skip connection} \\ \text{(= token embedding)}}} + \sum_{j=0}^{3} \underbrace{\text{attn\_out}^{(j)}}_{\substack{\text{attention head } j \\ \text{output (degree-2 trig)}}} \tag{9}$$

$$\text{pre}_n = \underbrace{W_{\text{in}}[n,:] \cdot \mathbf{x}^{(1)}}_{\text{linear projection of neuron } n} + \underbrace{b_{\text{in}}[n]}_{\text{bias}} \tag{10}$$

$$\text{MLP}[n] = \underbrace{\text{ReLU}\big(\text{pre}_n\big)}_{\substack{\text{activation of neuron } n \\ \approx \text{ degree-2 polynomial of } \cos(\omega_k a), \sin(\omega_k a), \ldots}} \tag{11}$$

$$\tag{12}$$

The core computation: For each key frequency k, the MLP neurons collectively compute:

$$\underbrace{\mathbf{u}_k^\top \cdot \text{MLP}(a,b)}_{\substack{\text{projection of MLP activations} \\ \text{onto direction } \mathbf{u}_k \text{ in } W_L}} \approx \underbrace{\alpha_k \cos(\omega_k(a+b))}_{\text{target: cosine of sum}} \tag{13}$$

$$= \overbrace{\underbrace{\alpha_k \cos(\omega_k a)\cos(\omega_k b)}_{\substack{\text{from neurons computing} \\ \cos(\omega_k a)\cos(\omega_k b)}} - \underbrace{\alpha_k \sin(\omega_k a)\sin(\omega_k b)}_{\substack{\text{from neurons computing} \\ \sin(\omega_k a)\sin(\omega_k b)}}}^{\text{cosine addition formula}} \tag{14}$$

$$\underbrace{\mathbf{v}_k^\top \cdot \text{MLP}(a,b)}_{\substack{\text{projection onto} \\ \text{direction } \mathbf{v}_k \text{ in } W_L}} \approx \underbrace{\beta_k \sin(\omega_k(a+b))}_{\text{target: sine of sum}} \tag{15}$$

$$= \overbrace{\underbrace{\beta_k \sin(\omega_k a)\cos(\omega_k b)}_{\substack{\text{from neurons computing} \\ \sin(\omega_k a)\cos(\omega_k b)}} + \underbrace{\beta_k \cos(\omega_k a)\sin(\omega_k b)}_{\substack{\text{from neurons computing} \\ \cos(\omega_k a)\sin(\omega_k b)}}}^{\text{sine addition formula}} \tag{16}$$

where $\mathbf{u}_k, \mathbf{v}_k \in \mathbb{R}^{512}$ are the directions in the neuron-logit map $W_L = W_{\text{out}}^\top W_U^\top$ corresponding to $\cos(\omega_k c)$ and $\sin(\omega_k c)$ respectively.

1.4 Step 4: Unembedding (Fourier → Logits)

The neuron-logit map $W_L = W_{\text{out}}^\top \cdot W_U^\top$ converts the MLP activations to output logits:

$$W_L \approx \sum_{k \in \mathcal{K}} \Big[\underbrace{\cos(\omega_k c)}_{\substack{\text{vector in } \mathbb{R}^P \\ \text{(c-th entry=}\cos(\omega_k c))}} \cdot \underbrace{\mathbf{u}_k^\top}_{\substack{\text{direction in neuron space} \\ \text{that reads } \cos(\omega_k(a+b))}} + \underbrace{\sin(\omega_k c)}_{\substack{\text{vector in } \mathbb{R}^P \\ \text{(c-th entry=}\sin(\omega_k c))}} \cdot \underbrace{\mathbf{v}_k^\top}_{\substack{\text{direction in neuron space} \\ \text{that reads } \sin(\omega_k(a+b))}} \Big] \tag{17}$$

$$\tag{18}$$

Therefore, the logit for output class c is:

$$\text{Logit}(c \mid a,b) = \underbrace{W_L \cdot \text{MLP}(a,b)}_{\text{neuron-logit map applied to MLP output}} \tag{19}$$

$$\approx \sum_{k\in\mathcal{K}} \Big[\underbrace{\cos(\omega_k c)}_{\substack{\text{from } W_L \\ \text{(unembed direction)}}} \cdot \underbrace{\mathbf{u}_k^\top \mathrm{MLP}(a,b)}_{\substack{\text{MLP computes this} \\ \approx\alpha_k \cos(\omega_k(a+b))}} + \underbrace{\sin(\omega_k c)}_{\substack{\text{from } W_L \\ \text{(unembed direction)}}} \cdot \underbrace{\mathbf{v}_k^\top \mathrm{MLP}(a,b)}_{\substack{\text{MLP computes this} \\ \approx\beta_k \sin(\omega_k(a+b))}} \Big] \tag{20}$$

$$\approx \sum_{k\in\mathcal{K}} \underbrace{\alpha_k \Big[\cos(\omega_k c)\cos(\omega_k(a+b)) + \sin(\omega_k c)\sin(\omega_k(a+b)) \Big]}_{\text{cosine subtraction identity}} \tag{21}$$

$$= \boxed{\sum_{k\in\mathcal{K}} \underbrace{\alpha_k \cos(\omega_k(a+b-c))}_{\substack{\text{peaks when } c\equiv a+b \pmod{113} \\ \text{since } \cos(0)=1 \text{ is maximum}}}} \tag{22}$$

1.5 Step 5: Prediction via Constructive Interference

$$\hat{c} = \underbrace{\arg\max_{c\in\{0,\dots,112\}}}_{\text{select class with highest logit}} \overbrace{\sum_{k\in\mathcal{K}} \alpha_k \cos\left(\frac{2\pi k(a+b-c)}{113}\right)}^{\text{sum of cosines at key frequencies}} \tag{23}$$

$$= \underbrace{(a+b) \mod 113}_{\substack{\text{constructive interference:} \\ \text{all cosines equal 1 when } c=a+b \bmod 113 \\ \text{destructive interference elsewhere}}} \tag{24}$$

1.6 Why Multiple Frequencies? (Constructive Interference)

A single cosine $\cos(\omega_k x)$ has period 113 but near-maxima at other values of x. By summing over multiple key frequencies, the model creates a function with a *unique* maximum at $x = 0 \bmod 113$:

$$f(x) = \sum_{k\in\mathcal{K}} \underbrace{\cos\left(\frac{2\pi k\cdot x}{113}\right)}_{\text{each has max at } x=0} \tag{25}$$

$$f(0) = \underbrace{3}_{\text{all 3 cosines=1}} \gg f(x\neq 0) \quad \underbrace{\text{(destructive interference)}}_{\text{cosines at different frequencies cancel}} \tag{26}$$

1.7 Worked Example: $a = 7, b = 13$

Step 1: Embed. Compute Fourier components for each key frequency k:

$$\begin{aligned}
k=3: &\quad \cos\left(\tfrac{2\pi\cdot 3\cdot 7}{113}\right), \sin\left(\tfrac{2\pi\cdot 3\cdot 7}{113}\right), \cos\left(\tfrac{2\pi\cdot 3\cdot 13}{113}\right), \sin\left(\tfrac{2\pi\cdot 3\cdot 13}{113}\right) \\
k=38: &\quad \cos\left(\tfrac{2\pi\cdot 38\cdot 7}{113}\right), \sin\left(\tfrac{2\pi\cdot 38\cdot 7}{113}\right), \cos\left(\tfrac{2\pi\cdot 38\cdot 13}{113}\right), \sin\left(\tfrac{2\pi\cdot 38\cdot 13}{113}\right) \\
k=51: &\quad \cos\left(\tfrac{2\pi\cdot 51\cdot 7}{113}\right), \sin\left(\tfrac{2\pi\cdot 51\cdot 7}{113}\right), \cos\left(\tfrac{2\pi\cdot 51\cdot 13}{113}\right), \sin\left(\tfrac{2\pi\cdot 51\cdot 13}{113}\right)
\end{aligned}$$

Step 2: MLP applies trig identity $\cos(\alpha)\cos(\beta) - \sin(\alpha)\sin(\beta) = \cos(\alpha+\beta)$:

$$k=3: \quad \underbrace{\cos\left(\tfrac{2\pi\cdot 3\cdot 7}{113}\right)\cos\left(\tfrac{2\pi\cdot 3\cdot 13}{113}\right) - \sin\left(\tfrac{2\pi\cdot 3\cdot 7}{113}\right)\sin\left(\tfrac{2\pi\cdot 3\cdot 13}{113}\right)}_{=\cos\left(\frac{2\pi\cdot 3\cdot 20}{113}\right)}$$

$$k=38: \quad \underbrace{\cos\left(\tfrac{2\pi\cdot 38\cdot 7}{113}\right)\cos\left(\tfrac{2\pi\cdot 38\cdot 13}{113}\right) - \sin\left(\tfrac{2\pi\cdot 38\cdot 7}{113}\right)\sin\left(\tfrac{2\pi\cdot 38\cdot 13}{113}\right)}_{=\cos\left(\frac{2\pi\cdot 38\cdot 20}{113}\right)}$$

Step 3: Logit for correct answer $c^* = (7+13) \bmod 113 = 20$:

$$\text{Logit}(20) = \sum_{k\in\mathcal{K}} \alpha_k \underbrace{\cos\left(\omega_k \cdot \overbrace{(7+13-20)}^{=0 \bmod 113}\right)}_{=\cos(0)=1} = \sum_k \alpha_k \quad \text{(MAXIMUM)}$$

For any $c \neq 20$, the cosines at different frequencies point in different directions and **destructively interfere**, giving a smaller logit.

1.7.1 Frequency $k = 3$: $\omega_3 = \frac{2\pi\cdot 3}{113}$

Embedding:

$$\underbrace{W_E[a]}_{\text{row } a \text{ of } W_E} \xrightarrow{\text{DFT}} \underbrace{\cos\left(\tfrac{2\pi\cdot 3\cdot a}{113}\right)}_{\text{component } k=3}, \quad \underbrace{\sin\left(\tfrac{2\pi\cdot 3\cdot a}{113}\right)}_{\text{component } k=3}$$

Attention:

$$\underbrace{A_0^{(j)} \approx 0.5 + \gamma_j \cos(\omega_3 a)}_{\text{head } j \text{ tuned to freq 3}} \times \underbrace{\text{OV}^{(j)}(a) \approx \cos(\omega_3 a)}_{\text{OV output at freq 3}}$$
$$\Rightarrow \underbrace{\cos^2(\omega_3 a),\ \cos(\omega_3 a)\cos(\omega_3 b),\ \ldots}_{\text{degree-2 products from attn}\times\text{OV}}$$

MLP (trig identity):

$$\underbrace{\text{neurons } n \in \text{cluster}_3}_{\text{1 neurons at freq 3}} \xrightarrow{\text{ReLU}}$$
$$\overbrace{\underbrace{\cos(\omega_3 a)\cos(\omega_3 b)}_{\text{from attn}} - \underbrace{\sin(\omega_3 a)\sin(\omega_3 b)}_{\text{from attn}}}^{=\cos(\omega_3(a+b))}$$

Unembedding:

$$\underbrace{\mathbf{u}_3^\top \cdot \text{MLP}}_{\text{project in } W_L} \approx \alpha_3 \cos(\omega_3(a+b))$$
$$\xrightarrow{\times\cos(\omega_3 c)} \underbrace{\alpha_3 \cos(\omega_3(a+b-c))}_{\text{freq 3 contribution to logit}(c)}$$

1.7.2 Frequency $k = 38$: $\omega_{38} = \frac{2\pi\cdot 38}{113}$

Embedding:

$$\underbrace{W_E[a]}_{\text{row } a \text{ of } W_E} \xrightarrow{\text{DFT}} \underbrace{\cos\left(\tfrac{2\pi\cdot 38\cdot a}{113}\right)}_{\text{component } k=38}, \quad \underbrace{\sin\left(\tfrac{2\pi\cdot 38\cdot a}{113}\right)}_{\text{component } k=38}$$

Attention:

$$\underbrace{A_0^{(j)} \approx 0.5 + \gamma_j \cos(\omega_{38} a)}_{\text{head } j \text{ tuned to freq 38}} \times \underbrace{\text{OV}^{(j)}(a) \approx \cos(\omega_{38} a)}_{\text{OV output at freq 38}}$$
$$\Rightarrow \underbrace{\cos^2(\omega_{38} a),\ \cos(\omega_{38} a)\cos(\omega_{38} b),\ \ldots}_{\text{degree-2 products from attn}\times\text{OV}}$$

MLP (trig identity):

$$\underbrace{\text{neurons } n \in \text{cluster}_{38}}_{\text{1 neurons at freq 38}} \xrightarrow{\text{ReLU}}$$

$$\overbrace{\underbrace{\cos(\omega_{38} a)\cos(\omega_{38} b)}_{\text{from attn}} - \underbrace{\sin(\omega_{38} a)\sin(\omega_{38} b)}_{\text{from attn}}}^{=\cos(\omega_{38}(a+b))}$$

Unembedding:

$$\underbrace{\mathbf{u}_{38}^{\top} \cdot \text{MLP}}_{\text{project in } W_L} \approx \alpha_{38} \cos(\omega_{38}(a+b))$$

$$\xrightarrow{\times \cos(\omega_{38} c)} \underbrace{\alpha_{38} \cos(\omega_{38}(a+b-c))}_{\text{freq 38 contribution to logit}(c)}$$

1.7.3 Frequency $k = 51$: $\omega_{51} = \frac{2\pi \cdot 51}{113}$

Embedding:

$$\underbrace{W_E[a]}_{\text{row } a \text{ of } W_E} \xrightarrow{\text{DFT}} \underbrace{\cos\left(\frac{2\pi \cdot 51 \cdot a}{113}\right)}_{\text{component } k=51}, \underbrace{\sin\left(\frac{2\pi \cdot 51 \cdot a}{113}\right)}_{\text{component } k=51}$$

Attention:

$$\underbrace{A_0^{(j)} \approx 0.5 + \gamma_j \cos(\omega_{51} a)}_{\text{head } j \text{ tuned to freq 51}} \times \underbrace{\text{OV}^{(j)}(a) \approx \cos(\omega_{51} a)}_{\text{OV output at freq 51}}$$

$$\Rightarrow \underbrace{\cos^2(\omega_{51} a),\ \cos(\omega_{51} a)\cos(\omega_{51} b),\ \ldots}_{\text{degree-2 products from attn}\times\text{OV}}$$

MLP (trig identity):

$$\underbrace{\text{neurons } n \in \text{cluster}_{51}}_{\text{0 neurons at freq 51}} \xrightarrow{\text{ReLU}}$$

$$\overbrace{\underbrace{\cos(\omega_{51} a)\cos(\omega_{51} b)}_{\text{from attn}} - \underbrace{\sin(\omega_{51} a)\sin(\omega_{51} b)}_{\text{from attn}}}^{=\cos(\omega_{51}(a+b))}$$

Unembedding:

$$\underbrace{\mathbf{u}_{51}^{\top} \cdot \text{MLP}}_{\text{project in } W_L} \approx \alpha_{51} \cos(\omega_{51}(a+b))$$

$$\xrightarrow{\times \cos(\omega_{51} c)} \underbrace{\alpha_{51} \cos(\omega_{51}(a+b-c))}_{\text{freq 51 contribution to logit}(c)}$$